Project-Q Workbook - 3

30-Day Quantum Computing Challenge

DAY - 03

Bloch Sphere & State Representation

Why Do We Need State Representation?

Imagine someone says:

"This qubit is in a superposition."

A natural question is:

Where exactly is that qubit?

In classical computing, the answer is simple:

Bit = 0 or 1

But in quantum computing:

Qubit = Infinite possible combinations of $|0\rangle$ and $|1\rangle$

We need a way to describe:

- How much of $|0\rangle$ exists
- How much of $|1\rangle$ exists
- The relationship between them

This description is called:

Quantum State Representation

Think About a Compass

A compass needle can point:

- North
- South
- East
- West
- Or any direction in between

Similarly,

A qubit can exist in:

- $|0\rangle$
- $|1\rangle$
- Or any valid quantum combination between them

Instead of representing only two possibilities,
We represent the qubit as a point in a state space.

State Representation

What Is a Quantum State?

A quantum state describes everything we know about a qubit before measurement.

Mathematically:

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

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Where:

α = amplitude of $|0\rangle$

β = amplitude of $|1\rangle$

$$|0\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$|1\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Important Rule

The probabilities must add up to 1.

$|\alpha|^2 + |\beta|^2 = 1$

Why?

Because after measurement the qubit must become either:

$|0\rangle$ OR $|1\rangle$

There is no third outcome.

Example 1

$|\psi\rangle = |0\rangle$

Meaning:

Probability($|0\rangle$) = 100%

Probability($|1\rangle$) = 0%

Visual:

$|0\rangle$
$|1\rangle$

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Example 2

$|\psi\rangle = |1\rangle$

Meaning:

Probability($|0\rangle$) = 0%

Probability($|1\rangle$) = 100%

Visual:

$|0\rangle$
$|1\rangle$

Example 3

$|\psi\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$

Meaning:

Probability($|0\rangle$) = 50%

Probability($|1\rangle$) = 50%

Visual:

$|0\rangle$ █████
$|1\rangle$ █████

This is one of the most important quantum states.

It is called:

Equal Superposition

Why State Representation Matters

Quantum algorithms do not directly manipulate bits.

They manipulate: Quantum States

The entire quantum computation process is simply:

Initial State

↓

Apply Quantum Gates

↓

New State

↓

Measurement

↓

Result

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Understanding state representation means understanding what the computer is actually processing.

The Problem With State Vectors

The equation

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

is mathematically correct.

But it is difficult to visualize.

Humans understand pictures better than equations.

Scientists needed a visual model.

The solution:

Bloch Sphere

What Is the Bloch Sphere?

The Bloch Sphere is a 3D visualization of a single qubit.

Every valid pure qubit state corresponds to one point on the surface of the sphere.

Think of it as:

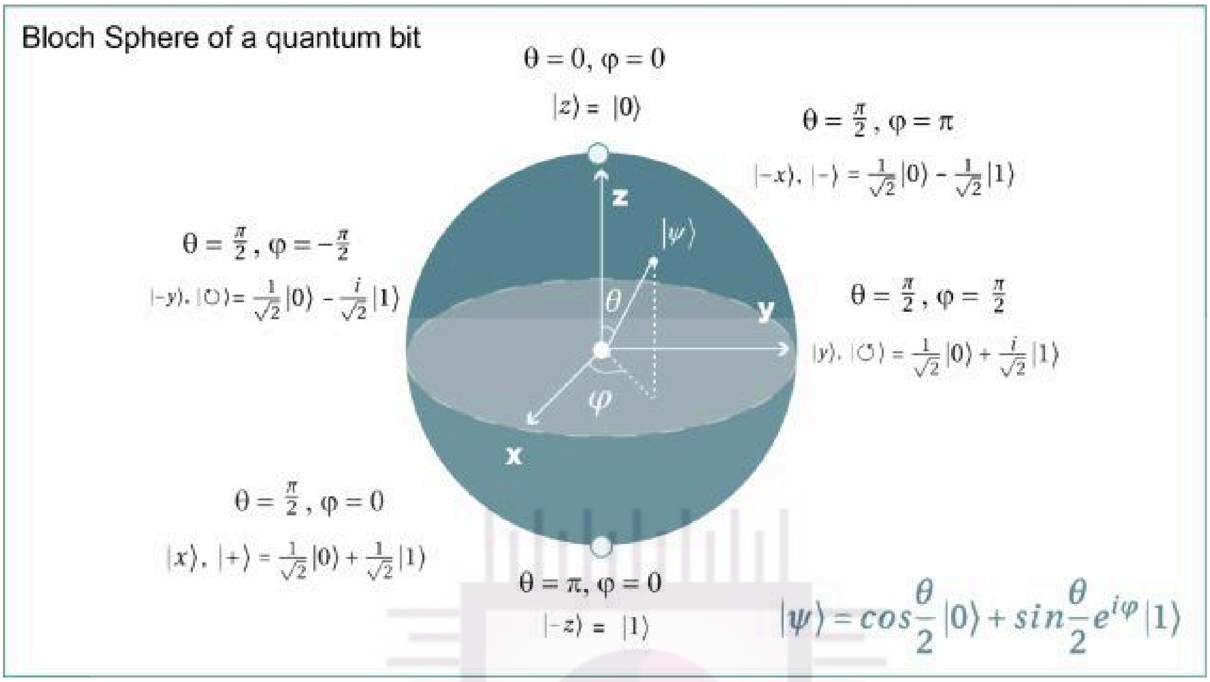
Google Maps for a qubit.

Instead of coordinates on Earth,

We use coordinates for quantum states.

Bloch Sphere Basics

Visual:



North Pole

Represents:

|0⟩

100% probability of measuring 0.

South Pole

Represents:

|1⟩

100% probability of measuring 1.

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Everywhere Else

Represents:

Superposition States

A qubit located between the poles contains information about both states simultaneously.

The Most Important States on the Bloch Sphere

Z-Basis States

North Pole → |0⟩

South Pole → |1⟩

These are computational basis states.

They are what we finally measure.

X-Basis States

Right Side → |+⟩

Left Side → |-⟩

Where:

$|+\rangle = (|0\rangle + |1\rangle)/\sqrt{2}$

$|-\rangle = (|0\rangle - |1\rangle)/\sqrt{2}$

These states are heavily used in quantum algorithms.

Y-Basis States

Front → |+i⟩

Back → |-i⟩

These states introduce phase information.

Phase is one of the key differences between classical and quantum computing.

How Quantum Gates Affect the Bloch Sphere

Think of a qubit as an arrow starting from the center.

Quantum gates rotate this arrow.

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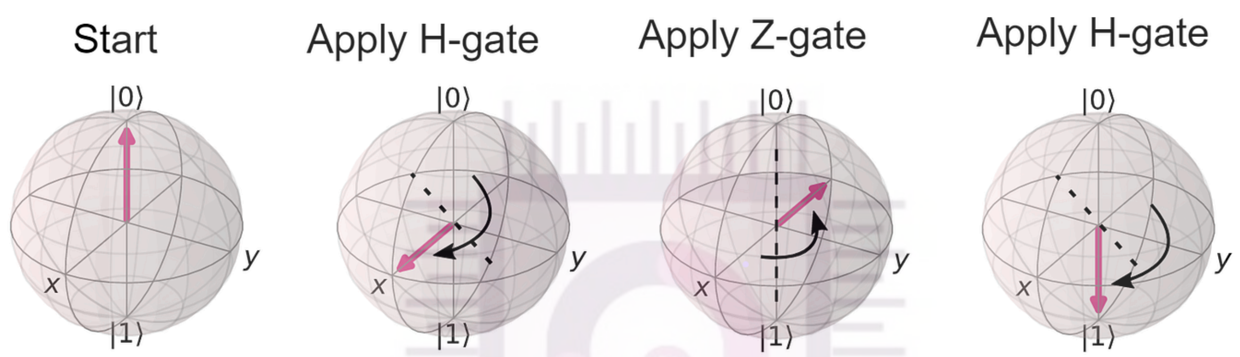
Visual:

Initial State
↓
Apply Gate
↓
New State

Quantum gates are essentially rotations of the qubit on the Bloch Sphere.



Why the Bloch Sphere Is Powerful

Without Bloch Sphere:

$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$

Looks abstract.

With Bloch Sphere:

You can literally see:

- State changes
- Superposition
- Rotations
- Gate operations
- Measurement directions

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This is why the Bloch Sphere is one of the most important visualization tools in quantum computing.

Limitation of the Bloch Sphere

The Bloch Sphere only works for:

One Qubit

When multiple qubits become entangled, the system can no longer be fully represented by a single Bloch Sphere.

So:

1 Qubit → Bloch Sphere

2+ Qubits → Need Advanced Representations

Real-World Analogy

Imagine Earth.

Every location on Earth can be identified using coordinates.

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Similarly:

Every pure qubit state can be identified using coordinates on the Bloch Sphere.

Earth Coordinate
↓

Latitude + Longitude

Bloch Sphere Coordinates
↓

Quantum Angles (θ, φ)

Quick Comparison

Concept	Purpose
State Representation	Mathematical description of a qubit
State Vector	Equation describing the qubit
Bloch Sphere	Visual representation of the qubit
Quantum Gates	Rotations on the Bloch Sphere
Measurement	Collapse to

End-of-Day Summary

What You Learned Today

✔ What a quantum state is

✔ State vector representation

✔ Amplitudes and probabilities

✔ Equal superposition state

✔ Bloch Sphere fundamentals

✔ Basis states (X, Y, Z)

✔ Quantum gate visualization

✔ Bloch Sphere limitations

✔ Relationship between states and measurement

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Top Verified Learning Resources

IBM Quantum (Highest Priority)

1. **Bloch Sphere Lesson (IBM Quantum Learning)**
   - https://quantum.cloud.ibm.com/learning/courses/general-formulation-of-quantum-information/density-matrices/bloch-sphere
   - Official IBM explanation used in quantum education. (IBM Quantum)

2. **Quantum State & Bloch Sphere Fundamentals**
    - https://quantum.cloud.ibm.com/learning/courses/utility-scale-quantum-computing/bits-gates-and-circuits
    - Excellent explanation of qubit states and visualization. (IBM Quantum)

Video Resources

1. **Mapping the Qubit State onto the Bloch Sphere**
    - Professor Nano (YouTube) (YouTube)
2. **The Bloch Sphere: Qubit Representation**
    - Quantum Leap (YouTube) (YouTube)

Additional Reference

3. **Quantum Education Modules – Bloch Sphere**
    - Beginner-friendly visual explanation with Qiskit examples. (Quantum Education Modules)

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